BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION SP/2024)

CLASS: B.TECH
BRANCH: CSE/IT

SEMESTER : VI
SESSION : SP/2024

SUBJECT: CS335 ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:
1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

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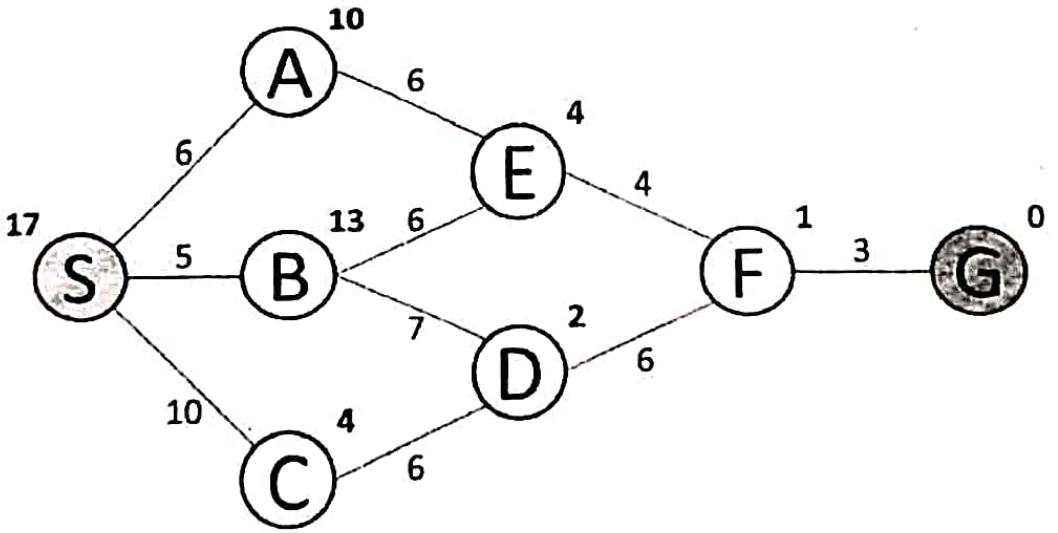
			CO	BL
Q.1(a)	Differentiate between uninformed search and informed search with an example.	[2]	1	4
Q.1(b)	What are limitations of Greedy best-first search algorithm? How are those limitations overcome by A* algorithm? Explain it with an example.	[3]	1	5
Q.2(a)	What is Heuristic search? Explain with an example.	[2]	1	1
Q.2(b)	Find the shortest path from node S to node G using A* algorithm. Consider flight-line distance as a heuristic function. The top value of each node denotes the flight-line distance of that node to node G. For example, the flight-line distance from node S to G and node A to G are 17 and 10 respectively. The value along with edges represents the actual distance between two nodes. For example, the actual distance between node A and node S is 6. Explicitly write down all the steps.	[3]	1	3



Q.3(a)	Give the name of the algorithm that results from each of the following special cases: (i) Simulated annealing with temperature T = 0 at all times (and omitting the termination test). (ii) Simulated annealing with Temperature T = ∞ at all times.	[2]	1	3
Q.3(b)	What are the limitations of the *hill climbing algorithm*? How can we overcome those limitations?	[3]	1	5
Q.4(a)	What is alpha-beta pruning?	[2]	1	1

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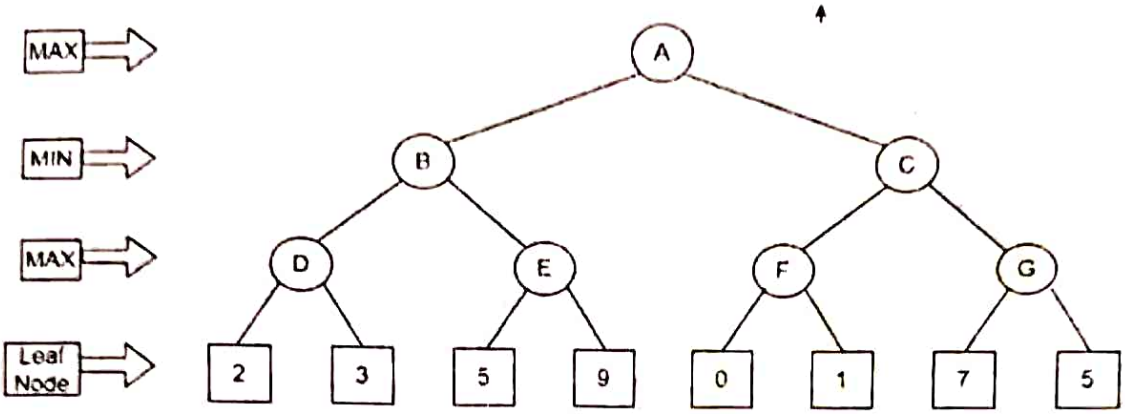
Q.4(b) Suppose, you are playing a game 'G' with your friend. In this game, your goal is to maximize your score, but your friend's goal is to minimize your score. Assume that each player has exactly 2 possible actions at every turn and both players are playing optimally. The below figure shows all the possible actions and utility values of the game 'G'. Find the optimal path of the game 'G' using the 'Mini-Max' algorithm with alpha-beta pruning. Explicitly write down all the steps. [3] 1 3

$$\alpha = -\infty, \beta = \infty$$



Q.5(a) What is knowledge representation in knowledge-based agent Artificial Intelligence system? Which kind of knowledge needs to represent in Artificial Intelligence system? [2] 2 1

Q.5(b) Consider the following sentences: [3] 2 3

(i) Ravi likes all kinds of food.

(ii) Apple and vegetables are food.

(iii) Anything anyone eats and not killed is food.

(iv) Ajay eats peanuts and is still alive.

(v) Rita eats everything that Ajay eats.

Converts these sentences into formulas in predicate logic (First order logic).

::::::21/02/2024::::::M